

Plecaz MR Tablet 60mg

Gliclazide

DESCRIPTION

15 mm length, white to off white uncoated oblong tablet, plain on one side and Pharmaniaga logo on the other side. PLECAZ MR 60 mg should not be split into half.

COMPOSITION

Each modified release tablet contains Gliclazide 60 mg.

PHARMACODYNAMICS

Gliclazide is a hypoglycaemic sulphonylurea oral antidiabetic active substances differing from other related compounds by an N-containing heterocyclic ring with an endocyclic bond.

Gliclazide reduces blood glucose levels by stimulating insulin secretion from the β -cells of the islets of Langerhans. Increase in postprandial insulin and C-peptide secretion persists after two years of treatment. In addition to these metabolic properties, gliclazide has haemovascular properties.

Effects in insulin release response to stimulation induced by a meal or glucose.

In type 2 diabetics, gliclazide restores the first peak of insulin secretion in response to glucose and increase the second phase of insulin secretion. A significant increase in insulin response is seen in response to stimulation induced by a meal or glucose.

Haemovascular properties

Gliclazide decreases microthrombosis by two mechanisms which may be involved in compilations of diabetes:

- a partial inhibition of platelet aggregation and adhesion, with a decrease in the markers of platelet activation (beta thromboglobulin, thromboxane B2).
- an action on the vascular endothelium fibrinolytic activity with an increase in tPA activity.

PHARMACOKINETICS

Absorption

Plasma levels increase progressively during the first 6 hours, reaching a plateau which is maintained from the 6 to 12 hours after administration. Intra-individual variability is low.

Gliclazide is completely absorbed. Food intake does not affect the rate or degree of absorption.

Distribution

Plasma protein binding is approximately 95%. The volume of distribution is around 30 litres. A single daily intake of Plecaz MR Tablet maintains effective gliclazide plasma concentrations over 24 hours.

Biotransformation

Gliclazide is mainly metabolised in the liver and excreted in the urine: less than 1% of the unchanged form is found in the urine. No active metabolites have been detected in plasma.

Elimination

The elimination half-life of gliclazide varies between 12 and 20 hours.

Linearity/non-linearity

The relationship between the dose administered ranging up to 120 mg and the area under the concentration time curve is linear.

Elderly

No clinically significant changes in pharmacokinetic parameters have been observed in elderly patients.

INDICATION

Non insulin dependent diabetes (type 2) in adults, when dietary measures, physical exercise and weight loss alone are not sufficient to control blood glucose.

RECOMMENDED DOSAGE

PLECAZ MR 60 mg is not able to deliver all approved dose regimens of gliclazide; other approved dosage forms and strengths of gliclazide should be used in such cases.

The daily dose of PLECAZ MR 60 mg may vary from one half to 2 tablets per day, i.e. from 30 to 120 mg taken orally in a single intake at breakfast time.

It is recommended to swallow the dose without crushing or chewing.

If a dose is forgotten, there must be no increase in the dose taken the next day.

As with any hypoglycaemic agent, the dose should be adjusted according to the individual patient's metabolic response (blood glucose, HbA1c).

Initial dose

The recommended starting dose is 30 mg daily. If blood glucose is effectively controlled, this dose may be used for maintenance treatment.

If blood glucose is not adequately controlled, the dose may be increased to 60, 90 or 120 mg daily, in successive steps. The interval between each dose increment should be at least 1 month except in patients whose blood glucose has not reduced after two weeks of treatment. In such cases, the dose may be increased at the end of the second week of treatment.

The maximum recommended daily dose is 120 mg.

• Switching from gliclazide 80 mg tablets to PLECAZ MR 60 mg modified release tablets:
One tablet of gliclazide 80 mg is comparable to 30 mg of the modified release formulation. Consequently the switch can be performed with careful blood monitoring.

• Switching from another oral antidiabetic agent to PLECAZ MR 60 mg:
PLECAZ MR 60 mg can be used to replace other oral antidiabetic agents.
The dosage and the half-life of the previous antidiabetic agent should be taken into account when switching to PLECAZ MR 60 mg.

A transitional period is not generally necessary. A starting dose of 30 mg should be used and this should be adjusted to suit the patient's blood glucose response, as described above.

When switching from a hypoglycaemic sulphonylureas with a prolonged half-life, a treatment free period of a few days may be necessary to avoid an additive effect of the two products, which might cause hypoglycaemia. The procedure described for initiating treatment should also be used when switching to treatment with PLECAZ MR 60 mg, i.e. a starting dose of 30 mg/day, followed by a stepwise increase in dose, depending on the metabolic response.

• Combination treatment with other antidiabetic agents:
PLECAZ MR 60 mg can be given in combination with biguanides, alpha glucosidase inhibitors or insulin.

In patients not adequately controlled with PLECAZ MR 60 mg, concomitant insulin therapy can be initiated under close medical supervision.

Special Populations

Elderly

PLECAZ MR 60 mg should be prescribed using the same dosing regimen recommended for patients under 65 years of age.

Renal impairment

In patients with mild to moderate renal insufficiency, the same dosing regimen can be used as in patients with normal renal function with careful patient monitoring. These data have been confirmed in clinical trials.

Patients at risk of hypoglycaemia:

- Undernourished or malnourished,
- Severe or poorly compensated endocrine disorders (hypopituitarism, hypothyroidism, adrenocorticotrophic insufficiency),
- Withdrawal of prolonged and/or high dose corticosteroid therapy,
- Severe vascular disease (severe coronary heart disease, severe carotid impairment, diffuse vascular disease);

It is recommended that the minimum daily starting dose of 30 mg is used.

Paediatric population

The safety and efficacy of PLECAZ MR 60 mg in children and adolescents have not been established. No data are available.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

The use of this medicine is contra-indicated in the following cases:

- hypersensitivity to gliclazide or other sulphonylureas or sulphonamides, or to any of the excipients used,
- insulin-dependent diabetes (type 1 diabetes),
- diabetic pre-coma and coma, diabetic keto-acidosis,
- severe hepatic or renal insufficiency; in this cases, the use of insulin is recommended
- treatment with miconazole
- lactation

WARNING AND PRECAUTIONS

Hypoglycaemia

This treatment should be prescribed only if the patient is likely to have a regular food intake (including breakfast). It is important to have a regular carbohydrate intake due to the increased risk of hypoglycaemia if a meal is taken late, if an inadequate amount of food is consumed or if the

food is low in carbohydrate. Hypoglycaemia is more likely to occur during low-calorie diets, following prolonged or strenuous exercise, alcohol intake or if a combination of hypoglycaemic agents is being used.

Hypoglycaemia may occur following administration of sulfonylureas. Some cases may be severe and prolonged. Hospitalisation may be necessary and glucose administration may need to be continued for several days.

Careful selection of patients, of the dose used, and clear patient directions are necessary to reduce the risk of hypoglycaemic episodes.

Factors which increase the risk of hypoglycaemia:

- patient refuses or (particularly in elderly patients) is unable to co-operate.
- malnutrition, irregular mealtimes, skipping meals, periods of fasting or dietary changes.
- imbalance between physical exercise and carbohydrate intake
- renal insufficiency
- severe hepatic insufficiency
- overdose of Plecaz MR
- certain endocrine disorders: thyroid disorders, hypopituitarism and adrenal insufficiency
- concomitant administration of certain other medicinal products

Renal and hepatic insufficiency

The pharmacokinetics and/or pharmacodynamics of gliclazide may be altered in patients with hepatic insufficiency or severe renal failure. A hypoglycaemic episode occurring in these patients may be prolonged, so appropriate management should be initiated.

Patient information

The risks of hypoglycaemia, along with its symptoms, treatment and conditions that predispose to its development, should be explained to the patient and to family members. The patient should be informed of the importance of following dietary advice, of taking regular exercise, and of regular monitoring of blood glucose levels.

Poor blood glucose control

Blood glucose control in a patient receiving anti-diabetic treatment may be affected by any of the following: St. John's Wort (*Hypericum perforatum*) preparations, fever, trauma, infection or surgical intervention. In some cases, it may be necessary to administer insulin.

The hypoglycaemic efficacy of any oral anti-diabetic agent, including gliclazide, is attenuated over time in many patients.

This may be due to progression in the severity of the diabetes, or to a reduced response to treatment. This phenomenon is known as secondary failure, which is distinct from primary failure, when an active substance is ineffective as first-line treatment. Adequate dose adjustment and dietary compliance should be considered before classifying the patient as secondary failure.

Dysglycaemia

Disturbances in blood glucose, including hypoglycaemia and hyperglycaemia have been reported, in diabetic patients receiving concomitant treatment with fluoroquinolones, especially in elderly patients. Indeed, careful monitoring of blood glucose is recommended in all patients receiving at the same time this medicine and a fluoroquinolone.

Laboratory tests

Measurement of glycated haemoglobin levels (or fasting venous plasma glucose) is recommended in assessing blood glucose control. Blood glucose self-monitoring may also be useful. Treatment of patients with glucose-6-phosphate (G6PD)-deficiency with sulfonylurea agents can lead to haemolytic anaemia. Since gliclazide belongs to the chemical class of sulfonylurea drugs, caution should be used in patients with G6PD-deficiency and a non-sulfonylurea alternative should be considered.

Porphyric patients

Cases of acute porphyria have been described with some other sulfonylurea drugs, in patients who have porphyria.

Excipients

Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Plecaz MR has no known influence on the ability to drive and use machines. However, patients should be made aware of the symptoms of hypoglycaemia and should be careful if driving or operating machinery, especially at the beginning of treatment.

INTERACTIONS WITH OTHER MEDICAMENTS

1) The following products are likely to increase the risk of hypoglycaemia:

Contra-indicated combination

Miconazole (systemic route, oral gel)

Increase in the hypoglycaemic effect with possible onset of hypoglycaemic symptoms, or even coma.

Combinations which are not recommended

Phenylbutazone (systemic route)

Increase in the hypoglycaemic effect of sulphonylureas (displacement of plasma protein binding and/or decrease in their elimination). It is preferable to use a different anti-inflammatory agent, or else to warn the patient and emphasise the importance of self-monitoring. Where necessary, adjust the dose during and after treatment with the anti-inflammatory agent.

Alcohol

Increases the hypoglycaemic reaction (by inhibiting compensatory reactions) that can lead to the onset of hypoglycaemic coma. Avoid alcohol or medicines containing alcohol.

Combinations requiring precautions for use

Potential of the blood glucose lowering effect and thus, in some instances, hypoglycaemia may occur when one of the following drugs is taken: Other antidiabetic agents (insulins, acarbose, metformin, thiazolidinediones, dipeptidyl peptidase-4 inhibitors, GLP-1 receptor agonist), beta-blockers, fluconazole, angiotensin converting enzyme inhibitors (captopril, enalapril), H₂-receptor antagonists, MAOIs, sulphonamides, clarithromycin and nonsteroidal anti-inflammatory agents.

2) The following products may cause an increase in blood glucose levels.

Combinations which are not recommended

Danazol (diabetogenic effect of Danazol)

If the use of this active substance cannot be avoided, warn the patient and emphasise the importance of urine and blood glucose monitoring. It may be necessary to adjust the dose of the anti-diabetic agent during and after treatment with danazol.

Combinations requiring special precautions during use

Chlorpromazine (neuroleptic agent)

High doses (>100 mg per day of chlorpromazine) increase blood glucose levels (reduced insulin release). Warn the patient and emphasise the importance of blood glucose monitoring. It may be necessary to adjust the dose of the anti-diabetic active substance during and after treatment with the neuroleptic agent.

Glucocorticoids (systemic and local route: intra-articular, cutaneous and rectal preparations) and tetracosactrin

Increase in blood glucose levels with possible ketosis (reduced tolerance to carbohydrates due to glucocorticoids). Warn the patient and emphasise the importance of blood glucose monitoring, particularly at the start of treatment. It may be necessary to adjust the dose of the antidiabetic active substance during and after treatment with glucocorticoids.

Ritodrine, salbutamol, terbutaline: (I.V.)

Increased blood glucose levels due to beta-2 agonist effects. Emphasise the importance of monitoring blood glucose levels. If necessary, switch to insulin.

Combination which must be taken into account

Anticoagulant therapy (e.g. Warfarin)

Sulfonylureas may lead to potentiation of anticoagulation during concurrent treatment. Adjustment of the anticoagulant may be necessary.

Saint John's Wort (*Hypericum perforatum*) preparations

Gliclazide exposure is decreased by Saint John's Wort-*Hypericum perforatum*. Emphasise the importance of blood glucose levels monitoring.

The following products may cause dysglycaemia:

Combinations requiring precautions during use

Fluoroquinolones

In case of a concomitant use of this medicine and a fluoroquinolone, the patient should be warned of the risk of dysglycaemia, and the importance of blood glucose monitoring should be emphasised.

PREGNANCY AND LACTATION

Pregnancy

There is no or limited amount of data (less than 300 pregnancy outcomes) from the use of gliclazide in pregnant women, even though there are few data with other sulfonylureas. As a precautionary measure, it is preferable to avoid the use of Gliclazide during pregnancy.

Control of diabetes should be obtained before the time of conception to reduce the risk of congenital abnormalities linked to uncontrolled diabetes.

Oral hypoglycaemic agents are not suitable, insulin is the drug of first choice for treatment of diabetes during pregnancy. It is recommended that oral hypoglycaemic therapy is changed to insulin before

a pregnancy is attempted, or as soon as pregnancy is discovered.

Lactation

It is not known whether gliclazide or its metabolites are excreted in breast milk. Given the risk of neonatal hypoglycaemia, the product is contraindicated in breast-feeding mother. A risk to the newborns/infants cannot be excluded.

ADVERSE EFFECTS

Based on the experience with gliclazide, the following adverse effects have been reported.

Hypoglycaemia

The most frequent adverse reaction with gliclazide is hypoglycaemia. As for other sulphonylureas, treatment with gliclazide tablets can cause hypoglycaemia, if mealtimes are irregular and, in particular, if meals are skipped. Possible symptoms of hypoglycaemia are: headache, intense hunger, nausea, vomiting, lassitude, sleep disorders, agitation, aggression, poor concentration, reduced awareness and slowed reactions, depression, confusion, visual and speech disorders, aphasia, tremor, paresis, sensory disorders, dizziness, feeling of powerlessness, loss of self-control, delirium, convulsions, shallow respiration, bradycardia, drowsiness and loss of consciousness, possibly resulting in coma and lethal outcome.

In addition, signs of adrenergic counter-regulation may be observed: sweating, clammy skin, anxiety, tachycardia, hypertension, palpitations, angina pectoris and cardiac arrhythmia. Usually, symptoms disappear after intake of carbohydrates (sugar). However, artificial sweeteners have no effect.

Experience with other sulphonylureas shows that hypoglycaemia can recur even when measures prove effective initially. If a hypoglycaemic episode is severe or prolonged, and even if it is temporarily controlled by intake of sugar, immediate medical treatment or even hospitalisation are required.

Other adverse effects

Gastrointestinal disturbances, including abdominal pain, nausea, vomiting dyspepsia, diarrhoea, and constipation have been reported: if these should occur, they can be avoided or minimised if gliclazide is taken with breakfast.

The following undesirable effects have been more rarely reported:

Skin and subcutaneous tissue disorders

Rash, pruritus, urticaria, angioedema, erythema, maculopapular rashes, and bullous reactions (such as Stevens-Johnson syndrome and toxic epidermal necrolysis) and exceptionally, drug rash with eosinophilia and systemic symptoms (DRESS).

Blood and lymphatic system disorders

Changes in haematology are rare. They may include anaemia, leukopenia, thrombocytopenia, granulocytopenia. These are in general reversible upon discontinuation of medication.

Hepato-biliary disorders

Raised hepatic enzyme levels (AST, ALT, alkaline phosphatase), hepatitis (isolated reports). Discontinue treatment in cholestatic jaundice appears. These symptoms usually disappear after discontinuation of treatment.

Eye disorders

Transient visual disturbances may occur especially on initiation of treatment, due to changes in blood glucose levels.

Class attribution effects

Cases of erythrocytopenia, agranulocytosis, haemolytic anaemia, pancytopenia, allergic vasculitis and hyponatraemia, have been described for other sulphonylureas. With other sulphonylureas cases were also observed of elevated liver enzyme levels and even impairment of liver function (e.g., with cholestasis and jaundice) and hepatitis which regressed after withdrawal of the sulphonyl urea or led to be life threatening liver failure in isolated cases.

OVERDOSE AND TREATMENT

An overdose of sulphonylureas may lead to hypoglycaemia. Moderate symptoms of hypoglycaemia, with no loss of consciousness or neurological signs, must be completely corrected by the administration of carbohydrates and by adjusting the dosages and/or dietary measures. Strict monitoring should be continued until the doctor is sure that the patient is out of danger.

Severe hypoglycaemic reactions, with coma, convulsions or other neurological disorder, are possible and constitute a medical emergency requiring the immediate hospitalisation of the patient.

If hypoglycaemic coma is diagnosed or suspected, the patient should be given intravenous (I.V)

injection of 50 mL of concentrated glucose solution (20to 30%). This should be followed by continuous infusion of a more dilute glucose solution (10%) at a rate that will maintain blood glucose levels above 1g/L. Patients should be monitored closely and depending on the patient's condition after this time, the doctor will decide if further monitoring is necessary.

Dialysis is of no benefit to patients due to the strong binding of gliclazide to proteins.

STORAGE CONDITION

Store below 30°C.

DOSAGE FORMS AND PACKAGING AVAILABLE

Box of 30 modified release tablets.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

**PRODUCT REGISTRATION HOLDER/
MANUFACTURER:
PHARMANIAGA MANUFACTURING BERHAD
(198001006232)**

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